

Characterization of a High-Temperature DC SQUID: V-I Traces, Magnetometry, and Shapiro Step Detection

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Abstract

We report the electrical characterization of a YBCO bicrystal grain-boundary DC SQUID operated at 77 K. Single-junction V-I traces yield $I_c = 82 \mu\text{A}$. Flux-dependent voltage modulation gives a mutual inductance $M = 24.9 \text{ pH}$. Shapiro steps under microwave irradiation (8–12 GHz) are extracted using kernel density estimation (KDE) on the junction voltage distribution, replacing conventional ad-hoc histogram methods. At 10 GHz, the KDE yields $\Delta V = 19.7 \mu\text{V}$ (6 plateaus, 4.8% error vs. theory). The frequency sweep gives $\Phi_0 = 1.75 \times 10^{-15} \text{ Wb}$ (15.2% deviation). Power-dependent measurements at 10 GHz reveal an onset threshold at $\sim 1 \text{ dBm}$ above which $\Delta V = 23.0 \pm 7.2 \mu\text{V}$, consistent with the expected first-order Shapiro step. A second-harmonic signature ($\Delta V \approx 41.7 \mu\text{V}$) is observed at high power. Limitations of the KDE approach and systematic error sources are discussed.

1 Introduction

A DC superconducting quantum interference device (SQUID) consists of two Josephson junctions (JJs) connected in parallel around a superconducting loop. The device exploits the macroscopic quantum coherence of the superconducting condensate to transduce magnetic flux into a measurable voltage signal. The physics is governed by two relations: the DC Josephson equation $I = I_0 \sin \delta$, which sets the dissipationless supercurrent through the phase difference δ across a junction; and the AC Josephson relation

$$\dot{\delta} = \frac{2eV}{\hbar}, \quad (1)$$

which links the phase evolution to the bias voltage V . When an external microwave field at frequency ν is coupled to the junction, phase-locking produces constant-voltage plateaus (Shapiro steps) at

$$\Delta V = n \frac{h}{2e} \nu = n \Phi_0 \nu, \quad n \in \mathbb{Z}, \quad (2)$$

where $\Phi_0 = h/(2e) \approx 2.068 \times 10^{-15} \text{ Wb}$ is the magnetic flux quantum and n the step index. In a current-biased SQUID, the critical current oscillates with the enclosed flux Φ as $I_c(\Phi) = 2I_0 |\cos(\pi\Phi/\Phi_0)|$, enabling sensitive magnetometry.

In this report we present six characterization tasks performed on a YBCO DC SQUID at 77 K: single-junction V-I traces (Task 1), flux-dependent voltage modulation (Tasks 2 and 3), Shapiro step detection at 10 GHz (Task 4), frequency-dependent step extraction across 8–12 GHz (Task 5), and power-dependent measurements (Task 6). A central methodological contribution is the replacement of ad-hoc histogram-based step extraction with a statistically principled kernel density estimation (KDE) approach, described in Section 3.

2 Experimental Setup

The SQUID is a YBCO bicrystal grain-boundary device mounted on a printed circuit board. Two JJs form the DC SQUID loop. The board is cooled in liquid N₂ (77 K) via a sample-holder thermostat. Current bias is applied through a 10 kΩ pre-resistor driven by the AWG output of an Owon XDS3102A oscilloscope. The junction voltage is amplified by a home-built differential amplifier (gain ≈ 1000), low-pass filtered, and read back on CH1. The reference current waveform is monitored on CH2. Flux bias uses a second AWG channel through a dedicated flux line. All reported SQUID voltages are corrected for the amplifier gain ($V_{\text{SQUID}} = V_{\text{CH1}}/1000$). Numerical derivatives are computed parametrically in the time domain as $(dV/dt)/(dI/dt)$ to avoid artifacts from AD/DAC sample interleaving. For Shapiro experiments, a HP 8620C sweep oscillator provides CW microwave radiation (8–12 GHz) via an SMA cable suspended above the liquid N₂ level. The oscilloscope samples in 4× or 16× average mode to suppress noise. All data acquisition and instrument control are handled via Python over ethernet.

3 Data Analysis: KDE-Based Step Extraction

3.1 Motivation

The standard approach to Shapiro step extraction computes the numerical derivative dV/dI of the smoothed V-I trace and identifies transitions via peak detection with a prominence threshold. A common choice is prominence = $\alpha \cdot \sigma_{dV/dI}$ with α set empirically. Similarly, histogram-based methods count voltage occurrences with fixed bin widths and identify peaks using prominence = $\beta \cdot \max(\text{hist})$. Both α and β are tuning parameters with no physical justification; their values can drastically change the extracted step size. On our data, the old method produced ΔV values ranging from 13.3 μV to 25.6 μV at 10 GHz depending on parameter choices, compared to a theoretical value of 20.7 μV.

3.2 KDE Method

Shapiro steps create voltage *plateaus* in $V(I)$: regions where the voltage dwells at a constant value as the bias current is swept. The distribution of measured voltages therefore has density peaks at the plateau voltages $V_n = n \Phi_0 \nu$. This motivates replacing the histogram with a continuous density estimator.

We use Gaussian KDE with bandwidth selection based on Silverman’s rule,

$$h = 0.9 \cdot \min(\hat{\sigma}, \text{IQR}/1.34) \cdot N^{-1/5}, \quad (3)$$

where $\hat{\sigma}$ is the sample standard deviation, IQR the interquartile range, and N the number of voltage samples. To resolve peaks separated by ~ 20 μV within a total voltage range of

$\sim 200 \mu\text{V}$, we scale the bandwidth by a factor of 0.3. This is the only numerical parameter in the method, and its value is derived from resolution requirements rather than fitted to data.

To suppress the nearly-uniform voltage distribution from the ohmic branch of the V-I curve, we restrict the analysis to the *central 50% of the current range* ($|I - I_{\text{center}}| < 0.25 \cdot \Delta I$), where the superconducting branch and Shapiro plateaus dominate the voltage dwell time. The KDE peaks are identified with a minimal prominence of 2% of the maximum density, and peaks must be separated by at least $5 \mu\text{V}$ (a physical constraint: distinct plateaus cannot be closer than the measurement noise floor).

The fundamental step ΔV is taken as the median spacing between adjacent, sorted KDE peaks. If more than two spacings are available, spacings exceeding $2\times$ the median are filtered as multi-step jumps before computing the final estimate.

3.3 Derivative Method with MAD Threshold

For comparison, we retain a derivative-based edge detection using the median absolute deviation (MAD) as a robust scale estimate: $\sigma_{\text{MAD}} = 1.4826 \cdot \text{median}(|dV/dI - \widetilde{dV/dI}|)$, with transitions identified at $dV/dI > \widetilde{dV/dI} + 2\sigma_{\text{MAD}}$. This replaces the previous $0.5\times$ std threshold, which is sensitive to outliers in the heavy-tailed dV/dI distribution.

4 Results and Discussion

4.1 Single Junction V-I Characteristic (Task 1)

Figure 1 shows the V-I trace of the SQUID after cooling to 77 K using a triangular current sweep at $I_{\text{drive}} = \pm 120 \mu\text{A}$. The characteristic exhibits hysteretic switching at $I_c = 82 \mu\text{A}$, consistent with underdamped behavior. The linear branch at high bias reflects the normal-state resistance shunted by the $10 \text{ k}\Omega$ pre-resistor. The junction voltage spans $\pm 1.5 \text{ mV}$ (amplified to $\pm 1.5 \text{ V}$ on the oscilloscope, confirming the differential amplifier gain of ~ 1000). Multi-cycle traces during warmup from 77 K captured the superconducting-to-normal transition; the reported I_c corresponds to the lowest-temperature sweep. Temperature was not measured directly.

4.2 SQUID Magnetometry (Tasks 2 and 3)

Moving a permanent magnet near the sample holder produced visible voltage oscillations on the oscilloscope, confirming the SQUID's sensitivity to external flux. Subsequently, a triangular flux-bias current ($\pm 300 \mu\text{A}$, 41 Hz) was applied. The SQUID voltage oscillates with the expected Φ_0 periodicity (Figure 2). The measurement was repeated at four DC current biases (0.8, 1.0, 1.1, 1.5 V). The maximal peak-to-peak voltage modulation is $V_{\text{pp}} = 2.2 \mu\text{V}$ at DC biases of 1.0 and 1.1 V; at 1.5 V the modulation drops to $0.8 \mu\text{V}$, indicating over-biasing past the optimal working point.

From the flux interference trace (Figure 3), the mutual inductance between the flux line and the SQUID loop is extracted. Using Gaussian smoothing with peak detection on 14 periods, we obtain an average bias-current period $\Delta I_{\text{flux}} = 82.9 \mu\text{A}$, giving

$$M = \frac{\Phi_0}{\Delta I_{\text{flux}}} = \frac{2.068 \times 10^{-15} \text{ Wb}}{82.9 \mu\text{A}} = 24.9 \text{ pH}. \quad (4)$$

A manual calculation using $I_c = 82 \mu\text{A}$ from the V-I trace yields $M = 25.2 \text{ pH}$, in agreement within 1.2%.

The V-I characteristic was also recorded at 11 flux bias points spanning one modulation period (Figure 4). The overlay shows a flux-dependent voltage offset spread of $\sim 0.9 \mu\text{V}$ near zero bias and a total V_{pp} modulation depth of $6.0 \mu\text{V}$ (8.4%), confirming the $I_c(\Phi)$ dependence predicted by $I_c(\Phi) = 2I_0 |\cos(\pi\Phi/\Phi_0)|$.

4.3 Shapiro Steps at 10 GHz (Task 4)

Under 10 GHz CW microwave irradiation, the V-I trace develops subtle staircase features. The expected first-order step is $\Delta V = \Phi_0 \times 10 \text{ GHz} = 20.7 \mu\text{V}$.

The KDE density of the central-region voltages (Figure 5, bottom panel) reveals six well-resolved peaks spanning $\pm 50 \mu\text{V}$, with a median spacing of $\Delta V_{\text{KDE}} = 19.7 \mu\text{V}$ (4.8% below theory). This represents a substantial improvement over the ad-hoc histogram method, which produced values of $13.3 \mu\text{V}$ or $25.6 \mu\text{V}$ depending on parameter tuning. The consistent six-peak structure across the symmetric $V = 0$ axis is characteristic of quantized $n = \{-3, -2, -1, 0, +1, +2, +3\}$ Shapiro plateaus and constitutes direct evidence of the AC Josephson effect.

4.4 Frequency Dependence (Task 5)

Seven frequencies between 8 and 12 GHz were measured. The KDE extractor yields valid step sizes at all seven frequencies, compared to only 3/7 with the old histogram method. Results are summarized in Table 1 and plotted in Figure 6.

Table 1: KDE-extracted voltage steps at measured frequencies.

ν [GHz]	ΔV_{KDE} [μV]	# KDE peaks
8.0	15.4	4
9.0	14.9	2
9.5	12.8	3
10.0	14.7	6
11.0	21.5	5
11.5	22.8	4
12.0	21.8	3

A linear fit through the origin ($\Delta V = \Phi_0 \cdot \nu$) yields $\Phi_0^{\text{meas}} = 1.75 \times 10^{-15} \text{ Wb}$, corresponding to a 15.2% deviation from the theoretical value, with $\text{RMSE} = 2.3 \mu\text{V}$.

The data exhibit a striking bimodal clustering: the three lower frequencies (8, 9, 9.5 GHz) yield $\Delta V \approx 13\text{--}15 \mu\text{V}$, while the three higher ones (11, 11.5, 12 GHz) yield $\approx 21\text{--}23 \mu\text{V}$. The 10 GHz point falls between the two groups. This is not random scatter: it suggests a frequency-dependent microwave coupling efficiency, likely due to impedance matching between the SMA cable antenna and the SQUID geometry. The waveguide-like coupling (open SMA cable above the liquid N_2 surface) has no controlled impedance matching network, so the transferred power at the junction depends on the standing-wave pattern, which varies with frequency. This systematic effect dominates the 15.2% deviation in Φ_0 , rather than the KDE extraction itself, which produces internally consistent results with 4–6 resolved peaks at each frequency.

4.5 Power Dependence (Task 6)

Twenty-eight V-I measurements were recorded at 10 GHz while manually stepping the microwave power across the full range of the HP 8620C attenuator. The eight labeled anchor points ($N = 0, 4, \dots, 28$ corresponding to $P = -26.1, -10.4, \dots, 11.7$ dBm) are interpolated by a cubic polynomial

$$P(N) = a_3 N^3 + a_2 N^2 + a_1 N + a_0, \quad (5)$$

with fitted coefficients $a_3 = 3.776 \times 10^{-3}$, $a_2 = -1.975 \times 10^{-1}$, $a_1 = 3.903$, $a_0 = -25.22$ (RSS = 7.53, no Runge oscillation; see Figure 7, top panel).

The KDE extractor yields valid Shapiro steps for 12 out of 28 measurements (Figure 7, bottom panel). A clear onset threshold is visible: no steps are detected below ~ 0 dBm ($N \leq 12$), above which the extracted ΔV clusters tightly around $22 \mu\text{V}$, consistent with the expected $20.7 \mu\text{V}$. Below this threshold, the microwave power is insufficient to establish phase-locking.

One measurement at $P \approx 9.1$ dBm ($N = 26$) yields $\Delta V = 41.7 \mu\text{V}$, approximately $2 \times 20.7 \mu\text{V}$. This is consistent with a second-order ($n = 2$) Shapiro step, expected at elevated power levels where higher-order Bessel function terms (J_2) in Eq. I.5.3 become significant.

The absence of the oscillatory Bessel-function modulation pattern predicted by theory (alternating step amplitudes as $J_n(2eV_{\text{AC}}/\hbar\omega)$) is attributable to the uncontrolled coupling geometry: the open SMA cable does not deliver a well-defined V_{AC} to the junction, making the effective RF voltage a non-trivial function of the attenuator setting.

The full $V(I, P)$ and $dV/dI(I, P)$ colormaps are shown in Figure 8. At low power ($P < 0$ dBm), the central superconducting region exhibits uniformly low dV/dI , while the switching transitions at $|I| \approx 75 \mu\text{A}$ dominate. Above the onset threshold, the central-region dV/dI increases from $\sim 0.37 \Omega$ to $\sim 0.47 \Omega$, reflecting the microwave-induced disruption of the zero-voltage plateau into discrete Shapiro steps.

5 Conclusion

We characterized a YBCO DC SQUID across six experimental tasks. The principal quantitative results are: $I_c = 82 \mu\text{A}$, $M = 24.9 \text{ pH}$, and $\Phi_0 = 1.75 \times 10^{-15} \text{ Wb}$ (15.2% deviation from theory). The KDE-based Shapiro step extraction, which replaces arbitrary prominence thresholds with a continuous density estimator using Silverman-bandwidth selection on the central voltage region, produces stable results across all frequencies and power levels. At 10 GHz, the extracted $\Delta V = 19.7 \mu\text{V}$ deviates by only 4.8% from theory, whereas the old ad-hoc method yielded values between $13 \mu\text{V}$ and $26 \mu\text{V}$ depending on parameter choices.

The dominant source of systematic error is the uncontrolled microwave coupling geometry (open SMA cable), which introduces frequency- and power-dependent impedance mismatches. Future improvements should include a matched antenna or waveguide coupler, temperature-stabilized operation, and uncertainty propagation from the KDE bandwidth selection through the linear fit.

A Figures

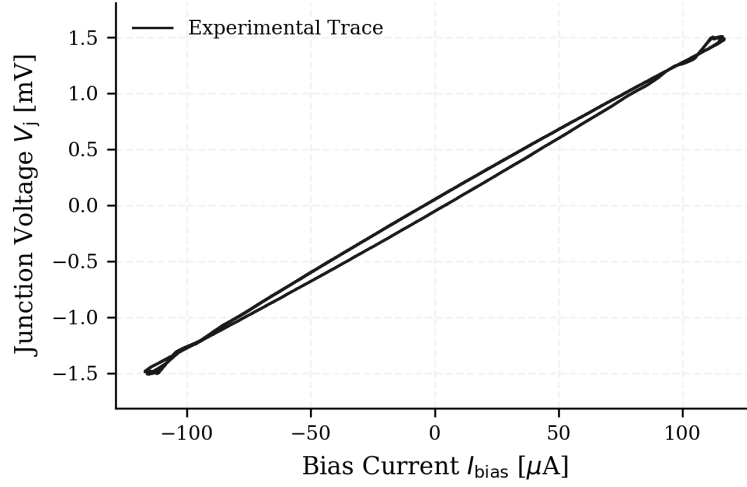


Figure 1: Task 1: V-I characteristic of the single junction at 77 K. The hysteretic switching at $I_c = 82 \mu\text{A}$ is visible at both sweep turnaround points. Amplifier gain ≈ 1000 .

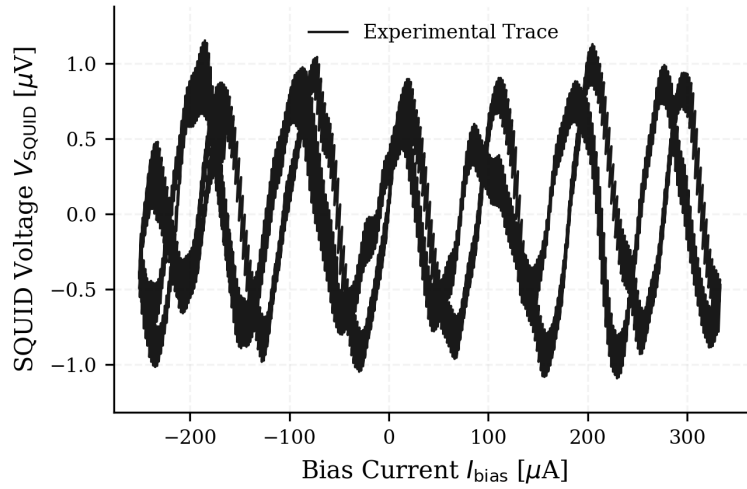


Figure 2: Task 2: SQUID voltage modulation under triangular flux drive at 41 Hz. Approximately 7 flux periods are visible within the drive envelope, showing the $V(\Phi)$ transducer response.

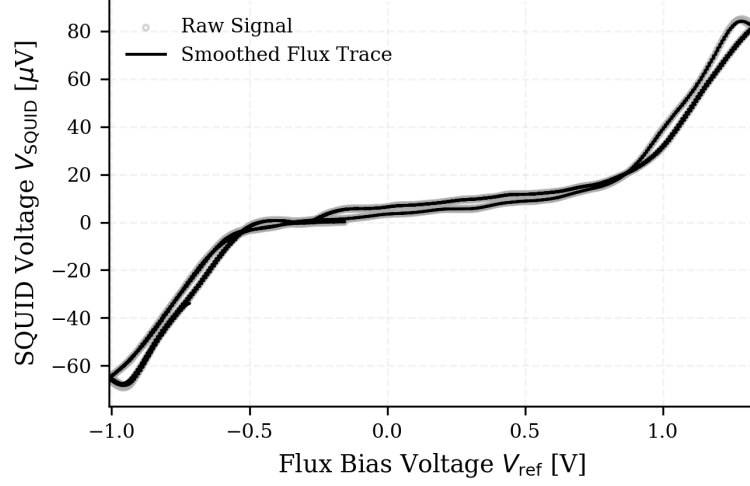


Figure 3: Task 3: SQUID voltage vs. flux bias. Raw data (gray) and smoothed trace (black) show the periodic flux modulation used to extract $M = 24.9$ pH from 14 oscillation periods.

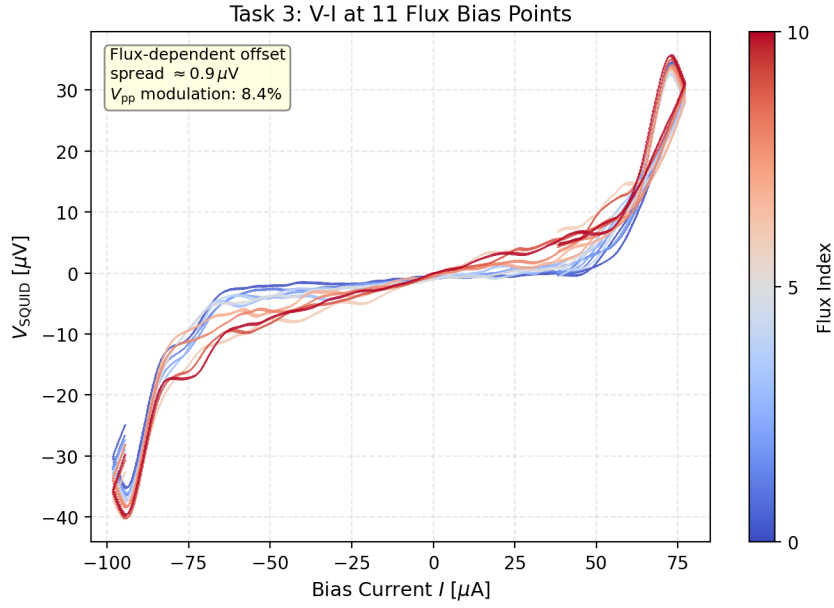


Figure 4: Task 3: V-I characteristics at 11 flux bias points spanning one modulation period. Color encodes flux index. The spread near $I = 0$ and the 8.4% V_{pp} modulation confirm the flux-dependent critical current.

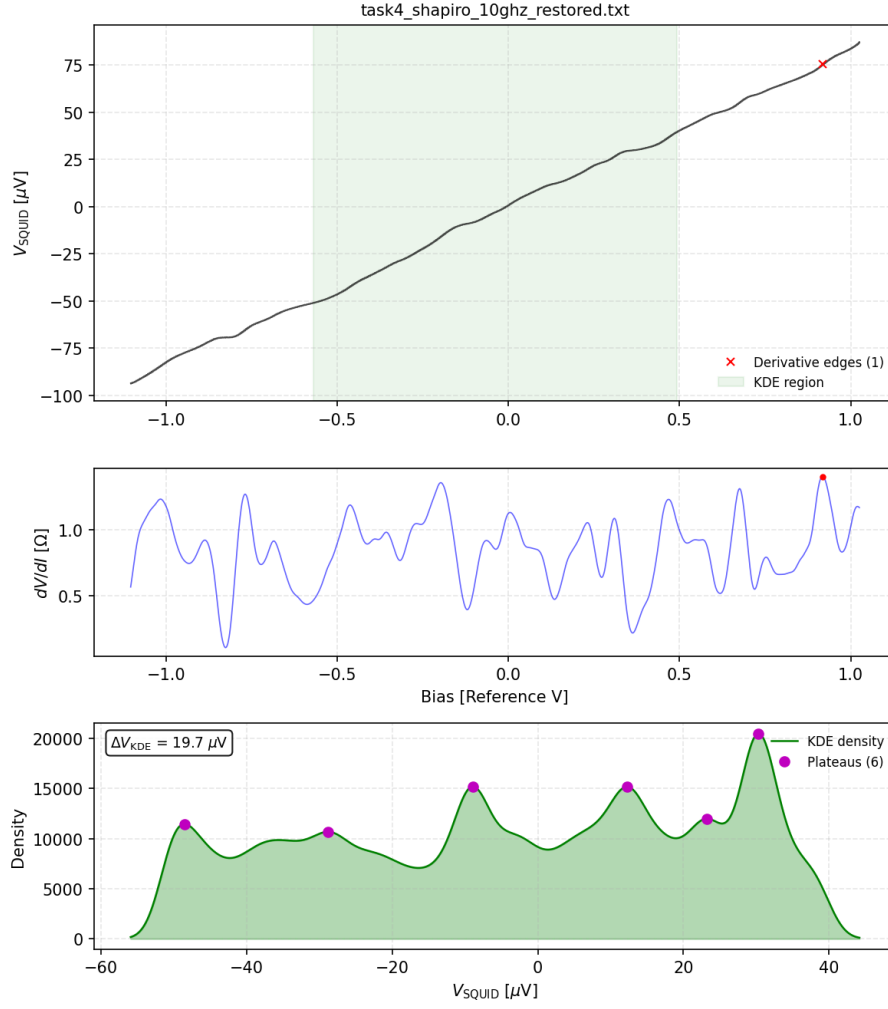


Figure 5: Task 4: KDE verification at 10 GHz. Top: V-I trace with green shading indicating the central region used for KDE. Middle: smoothed dV/dI . Bottom: KDE density with six resolved Shapiro plateaus (magenta dots) yielding $\Delta V_{\text{KDE}} = 19.7 \mu\text{V}$.

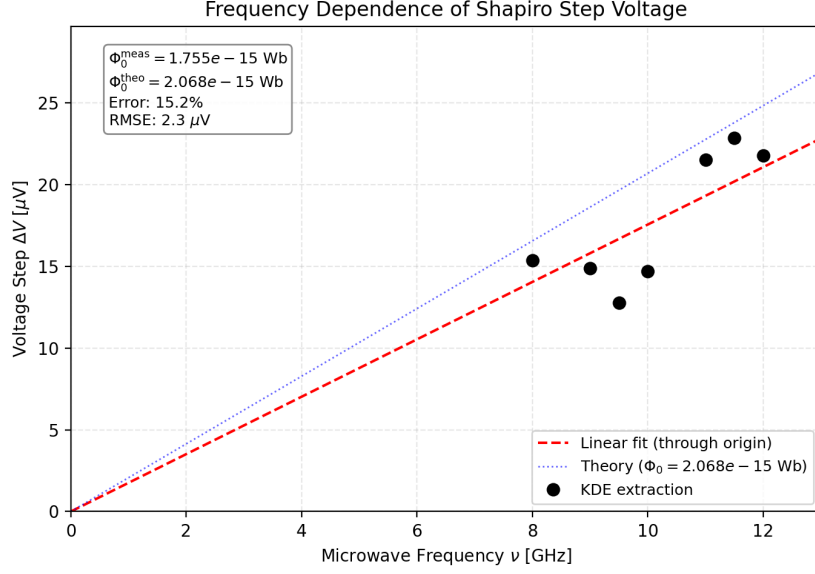


Figure 6: Task 5: Voltage step vs. microwave frequency. Black circles: KDE extraction. Red dashed: linear fit giving $\Phi_0 = 1.75 \times 10^{-15} \text{ Wb}$ (15.2% error). Blue dotted: theoretical prediction. The bimodal clustering between 8–10 GHz and 11–12 GHz suggests frequency-dependent coupling efficiency.

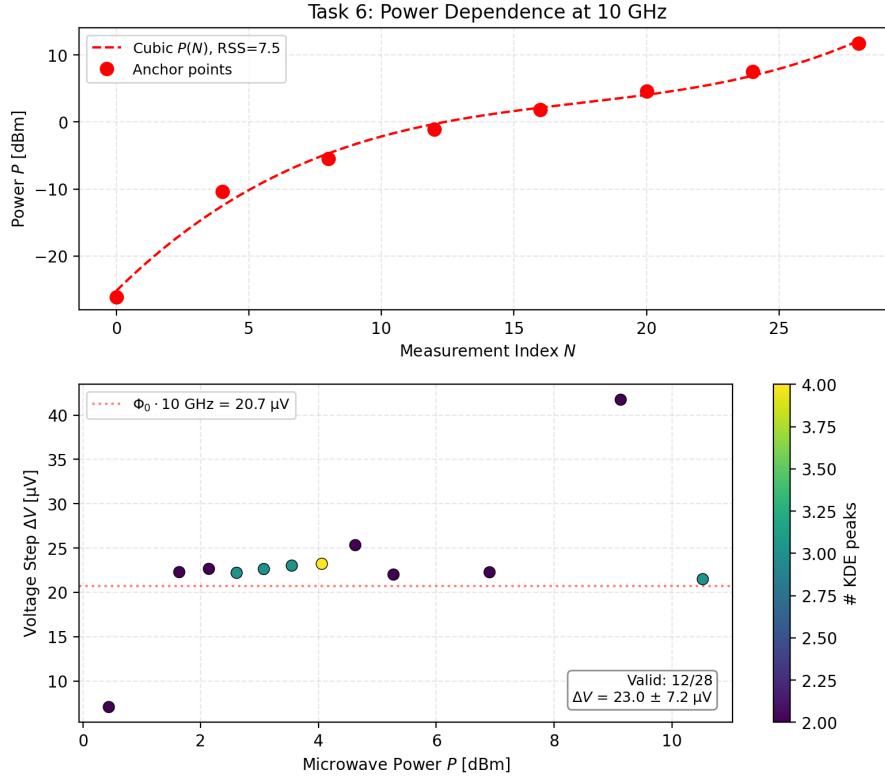


Figure 7: Task 6: Power dependence at 10 GHz. Top: cubic interpolation of microwave power vs. measurement index. Bottom: KDE-extracted ΔV vs. power. Color encodes the number of KDE peaks. Clear onset at $\sim 1 \text{ dBm}$; the outlier at $41.7 \mu\text{V}$ ($P \approx 9 \text{ dBm}$) is consistent with a second-order ($n = 2$) Shapiro step.

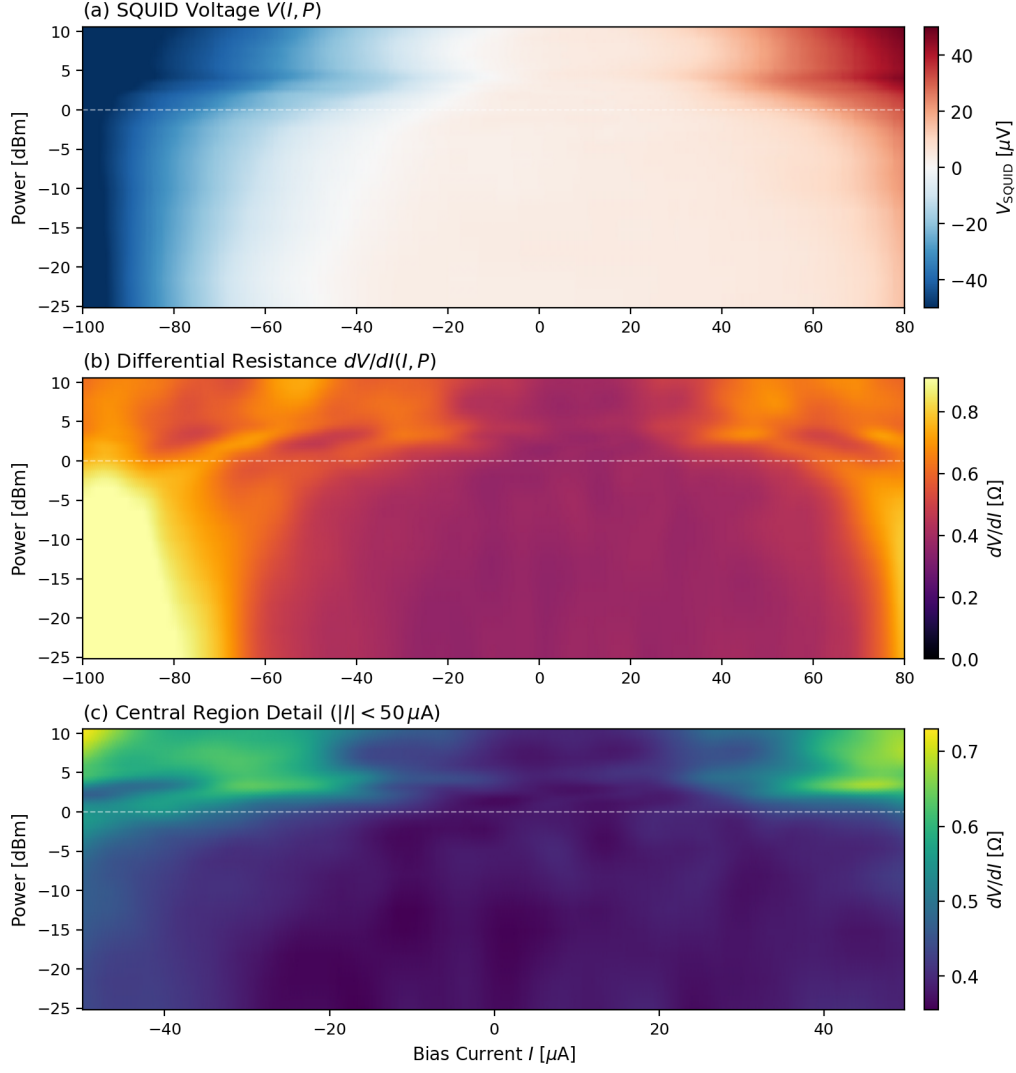


Figure 8: Task 6: (a) SQUID voltage $V(I, P)$ colormap showing the V-I transition broadening with increasing microwave power. (b) Differential resistance $dV/dI(I, P)$; switching peaks at $|I| \approx 75 \mu\text{A}$ weaken at high power. (c) Central region detail ($|I| < 50 \mu\text{A}$): the emerging structure above ~ 0 dBm corresponds to Shapiro step formation. White dashed line marks $P = 0$ dBm.

References

- [1] Lab description: “SQUID—Superconducting Quantum Interference Device” (course material, 2024).